

Wave Release Coordination under Vertical Resource Constraints in Multi-Story AMR Warehouse

Abstract

In multi-story fulfillment warehouses, autonomous mobile robots (AMRs) are floor-confined and share a few freight elevators, making elevator capacity the primary bottleneck. Wave composition (which orders to release together) largely fixes makespan before any AMR moves but has never been treated as a decision variable coupled to shared vertical transport. We formalize this wave-release coordination problem, fix operational dispatch, and group waves by three features: vertical spread, directional imbalance, and temporal clustering. Two tools using this grouping answer two operator questions. The Bound-and-Gap decomposition answers what wave composition is worth, splitting it into a structural ceiling no policy can recover and a recoverable slack equal to a predict-then-optimize loss that existing methods can close. A resolution theorem shows finer grouping can only raise the ceiling. The Model-Dominance Hedge Rule determines which structure to release when the elevator model is unknown by selecting the one performing best under the most conservative model. A dominance theorem (any elevator count, two failure channels) proves this simple hedge minimizes worst-case makespan and equals a calibrated, distributionally robust optimum. A pre-registered 106,800-simulation study tests both tools. The Hedge Rule passes every gate: dominance holds in 99.2% of matched waves in the six-configuration block, and its selection matches that optimum in all six. The decomposition identities hold exactly (72/72), with a partial resolution pass where gaps are small. The capacity-versus-policy verdict is seed-stable, unanimous in 15/18 configurations. The two tools inform operators when composing waves pays and when capacity is the lever.

Introduction

The growth of e-commerce and the demand for next-day delivery have pushed distribution centers into dense urban areas where land is scarce and expansion is constrained. Operators increasingly turn to multi-story fulfillment buildings: floors stacked vertically, with autonomous mobile robots (AMRs) handling horizontal movement on each floor and a few freight elevators connecting them. The primary bottleneck has shifted from floor space or fleet size to elevator capacity, and AMRs often idle while waiting for an elevator.

Operators relieve this congestion by releasing orders in waves: within a short release window, a selected subset of orders is dispatched simultaneously. The composition of each wave, that is, which orders to bundle, directly determines how many vertical transitions the elevators must serve, in which directions, and with what temporal stagger. This determines the difficulty of the dispatch problem before any AMR moves. In current practice, wave composition is set heuristically, by destination floor, carrier cutoff, or arrival order. However, the wave layer is the cheapest place to relieve a vertical bottleneck, because re-batching already-released orders changes neither the layout nor the fleet size.

Four elements, namely wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity, have largely been studied in isolation. The closest precedents treat the lift as a constraint under fixed wave content (Chakravarty et al., 2025) or batch orders inside a single vertical lift module (VLM) (Lenoble et al., 2018). Multi-story Robotic Mobile Fulfillment System (RMFS)

work avoids the coupling by confining shuttles to a tier (J. Wu et al., 2025). Planar AMR work optimizes wave cardinality on a single floor (Qin et al., 2024). Each of these precedents sidesteps the coupling by restricting the setting: fixing the wave content, or confining transport to a single device or a single tier. In the general setting we study, however, a flexible AMR fleet shares building-wide elevators and wave composition is itself a decision. The two choices then cannot be separated: the chosen wave fixes the elevator trip distribution (how many vertical transitions, in which directions, with what stagger) before any AMR is dispatched, so optimizing wave composition and fleet dispatch in isolation loses the bottleneck interaction that matters. We therefore formalize the intersection of these four elements as the wave-release coordination problem under vertical resource constraints; Section 2 details the precedents.

The problem decomposes naturally into two stages: a tactical stage composes each wave, and an operational stage dispatches the AMRs that deliver its orders through the shared elevators. We hold the operational stage fixed and study the tactical stage with two complementary analytical tools, both built on a three-dimensional representation $\Phi = (C, I, T)$ of each wave: its vertical spread C , directional imbalance I , and temporal clustering T . Two structural features motivate the tools. First, the value that wave structure can deliver varies across operating regimes. The shared elevator is the bottleneck: in some regimes makespan is controlled by the elevator's capacity and can be relieved only by adding capacity, while in others it depends on how waves are composed, so the wave-release policy can relieve it. The value therefore has two parts. One is fixed by capacity, and no wave-release policy can recover it. The other is slack that a Φ -informed wave-release policy can recover. Which part dominates shifts across regimes, so the value of composing waves must be measured before it is pursued. Second, the operator does not know which elevator model is true, but on almost every wave the deterministic candidate models rank the makespans in the same order. Among them the most conservative, true co-occupancy batching, is weakly the slowest and serves as a worst-case reference. This ordering holds wave by wave, not only on average. The operator can therefore score the candidate waves against that single reference and release the one that minimizes makespan under it. That choice is robust: we prove it minimizes the worst-case makespan across the candidate models and coincides with a calibrated distributionally-robust optimum, with a computable worst-case-loss bound. The two tools act on the same decision (which wave-structure to release): the first measures how much that choice is worth and how much of it a wave-release policy can recover, and the second uses the reference model to select the robust one.

This paper makes three contributions.

(C1) Problem formulation. We formalize wave-release coordination under vertical resource constraints as a two-stage scheduler: a tactical stage composes waves; an operational stage dispatches a flexible AMR fleet through shared freight elevators. Its four defining elements (wave composition as a decision, multi-story deployment, a flexible AMR fleet, and shared elevator capacity) have been studied only in isolation, with wave content or request set fixed or transport confined to a single device or tier; our formulation models the coupling explicitly.

(C2) Methodology. We develop two analytical tools on the wave representation $\Phi = (C, I, T)$. The first, the Bound-and-Gap decomposition, is diagnostic. It splits the value of wave structure, GAP ,

into a structural ceiling H_{up} that no wave-release policy on Φ can recover and a recoverable slack M_Φ (Proposition 1). We show that M_Φ equals the predict-then-optimize decision loss of a partition-constant predictor (Proposition 3; Adam N. Elmachtoub & Paul Grigas (2021)), so the slack can be recovered with existing methods. A resolution theorem shows that grouping waves more finely can only raise the ceiling (Theorem R), and Corollary 1 applies this to the specific wave classes used in our experiments. The second tool, the Model-Dominance Hedge Rule, is prescriptive. It selects the wave-structure to release in closed form, without identifying the true elevator model. Its guarantee rests on Proposition 2, which proves this wave-by-wave ordering, called chain dominance, under verifiable conditions and for any number of elevators. The proof also identifies the only two ways the ordering can fail. Under this dominance, minimax corner selection collapses to the corner optimal under true co-occupancy batching (Theorem 1) and coincides with the corner-calibrated Wasserstein distributionally robust optimum (Corollary 3; Mohajerin Esfahani & Kuhn (2018)). When dominance holds only approximately, Corollary 2 provides a computable worst-case bound $U_{c(\epsilon)}$.

(C3) Empirical capacity-versus-policy diagnosis. For an operator, the practical question is whether makespan is better improved by composing waves more smartly or by adding elevator capacity. Our main finding is that this question has a stable, measurable answer. In these warehouses, under the pre-registered wave grouping and the median metric, capacity is the binding side. About 72% of the mean value of wave composition (GAP) sits in the structural ceiling that no release policy can recover, leaving about 28% as slack that a better release policy can still claim. This result is consistent across random seeds and is unanimous in 15 out of 18 configurations. The 72% is a measured share, not a universal constant. It changes when waves are grouped more finely or summarized with different statistics, and Theorem R makes this dependence predictable rather than arbitrary. These findings come from a pre-registered publication-scale study of 18 configurations and 106,800 simulations. The Hedge Rule passes every pre-registered test. The ordering it relies on holds in 99.2% of the matched wave pairs in the six configurations where the models are compared, never dropping below 96.0% in any single cell, and persists as elevator capacity varies from 2 to 5 loads per trip. In all six of those configurations, the wave class the rule selects is the one a full distributionally-robust optimization would select. The decomposition's two identities hold exactly in all 72 experimental cells. Its statistical test resolves a nonzero value of wave composition in 57 of the 72 cells, which we report as a partial pass. The unresolved cells are precisely those where the value is smallest (discriminating power, AUC 0.92), so the decomposition detects value wherever there is meaningful value to detect. Together, the tools tell an operator, regime by regime, when smarter wave composition pays and when the lever is instead more capacity or a finer grouping.

Section 2 reviews the related literature. Section 3 formulates the problem. Section 4 develops the two tools. Section 5 presents computational experiments. Section 6 concludes.

Related works

Our work sits at the intersection of several established bodies of literature, each touching a part of the problem without addressing its joint form. We position the contribution in two steps: recent surveys frame the warehouse-OR landscape, then four problem-class threads whose intersection

defines our gap, and finally three methodological threads (two bridges and one contrast). Recent surveys frame the landscape: Boysen & de Koster (2025) give a fifty-year, three-generation classification that places robotized distribution centers as the current generation; Boysen et al. (2019) survey warehousing in the e-commerce era; Azadeh et al. (2019) review robotized and automated warehouse systems; and Pardo et al. (2024) provide the most recent taxonomy of the order-batching family. Within the scope of these surveys, we do not find wave-release coordination under shared, multi-story elevator coupling treated as a standalone problem class, consistent with our positioning this work as a formalization step.

Treating the composition of an order batch as a decision variable, not merely its size, has a long tradition in warehouse OR, but almost entirely in planar or device-level settings. Gademann et al. (2001) study wave picking with a makespan objective in parallel-aisle warehouses; Bozer & Kile (2008) and the canonical treatment of Bartholdi & Hackman (2019) embed the decision in walk-and-pick systems; Ardjmand et al. (2018) minimize order-picking makespan with multiple pickers (our objective, but single-floor and without a shared vertical resource); and Rasmi et al. (2022), Schiffer et al. (2022) and Haouassi et al. (2022) extend wave and order-line batching in e-commerce settings. AMR-assisted variants enlarge the decision class: Scholz et al. (2017) jointly solve order batching, batch assignment and picker routing; Žulj et al. (2022) lift it to AMR-assisted picker-to-parts systems; and Qin et al. (2024) study a single-layer multi-tote system, use the word “wave” for processing batches, and identify an optimal wave cardinality. At the device level, Lenoble et al. (2018) study order batching across several vertical lift modules, and Boysen et al. (2018) the minimum order-spread sequencing problem in automated sorting; the mathematics in both is internal to a single device and does not generalize to multi-elevator capacity shared across a horizontal AMR fleet. Recent RMFS batching work (e.g., Liu et al. (2025)) likewise targets single-tier picking throughput rather than vertical coupling. Our delta is to represent wave composition through Φ in a multi-story setting so that it shapes the elevator trip distribution; remove the multi-story dimension or the shared elevator and our problem collapses to their planar subclass.

Multi-story Robotic Mobile Fulfillment Systems (RMFS) and multi-tier shuttle systems make vertical transport explicit, but most decouple tiers by confining shuttles or pods to a tier, or by modeling the lift as an exogenous queue, with the task set given. Z. Wu et al. (2024) schedule inbound jobs in a four-way shuttle system; Tadumadze et al. (2023) assign orders and pods to picking stations in a multi-level RMFS; Lamballais et al. (2017) give queueing estimates of RMFS performance. On the shuttle-based storage-and-retrieval (SBS/RS) side, Tappia et al. (2015) provide the canonical queueing model of multi-tier compact storage with lifts, and Chen et al. (2023) schedule retrieval requests under two lifts to minimize makespan (the closest published analogue to ours, but operational, with the task set given, and for SBS/RS rather than floor-bound AMRs); more recent integrated work (J. Wu et al., 2025) still optimizes picking and replenishment within a single tier. A recent exception to tier confinement is Wang et al. (2025), who schedule retrievals in a four-directional shuttle-based compact storage and retrieval system whose tier-to-tier shuttles traverse tiers via two shared, heterogeneous lift types (shuttle lifts and bin lifts), jointly deciding request sequencing and the coordinated shuttle and lift schedules that minimize makespan, combining logic-based Benders decomposition with adaptive large neighborhood search and tabu search. The decision layer, however, is unchanged: the work remains at the operational stage, its retrieval-request set is exogenous, no

wave-composition decision exists upstream, and its tier-to-tier shuttles are storage-and-retrieval devices rather than a floor-bound AMR fleet contending for building-level freight elevators. Our novelty is to treat the cross-tier lift coupling as something an upstream tactical decision that can be managed by wave composition rather than as a contention that must be rescheduled after the request set is finalized.

The precedents closest to our physical setting treat the elevator as a contended constraint under fixed wave content. Chakravarty et al. (2025) compute provably optimal lift schedules for AMR fleets via Boolean satisfiability, with an exogenous task set; Tsai et al. (2025) optimize elevator standby strategies in smart buildings; and Richer et al. (2024) give an OR treatment of static elevator dispatching under destination control. These works optimize lift dispatch given an arrival process; we optimize the wave that generates it. A parallel line controls warehouse fleets through learning rather than formal optimization: Crites et al. (1998) established the reinforcement-learning treatment of elevator group control, and the line continues today (traffic-pattern-aware deep RL dispatching, Wan et al. (2024); deep RL for batch-order scheduling in RMFS, Cheng et al. (2024); multi-robot task allocation, Ma et al. (2025), Dhanaraj et al. (2025), Wen & Ma (2024); and an RL-for-AMR -fleets survey, Wesselhöft et al. (2022)). These methods learn dispatch end-to-end at the operational layer and answer a different question from ours: they expose neither the Bound-and-Gap structural-versus-recoverable split nor a training-free closed-form reference, so we regard them as complementary rather than competing. Finally, a note on sign: the classic elevator round-trip-time (RTT) literature shows that batching same-destination passengers reduces stops and round-trip time (Al-Sharif et al., 2014; So et al. (2022)), that is, an individual-rider model overstates travel time, whereas our chain dominance runs the opposite way (true co-occupancy batching is weakly slower than the throughput abstraction); the sign follows from two assumptions of our setting, detailed in Section 4.

Three methodological threads ground our two tools; the first two are connected to them by precise equivalences in Section 4, the third by a structural contrast. First, the prediction-to-decision regret literature bounds the loss when a learned predictor drives a downstream optimizer (Adam N. Elmachtoub & Paul Grigas, 2021; Chenreddy & Delage, 2024; Vera et al., 2020), with recent decision-focused -learning surveys (Mandi et al., 2024) and predict-then-optimize generalization bounds (Balghiti et al., 2022) characterizing the learnability of that loss. Proposition 3 gives this loss a partition perspective: the policy slack M_Φ of our decomposition equals the SPO loss of a partition-constant predictor, so existing decision-focused methods can recover it, stated as a decision loss, not as a claim that Φ predicts makespan. The resolution behavior of the decomposition draws on a second classical idea, that refining a partition can only sharpen what it reveals, familiar from Blackwell's comparison of experiments (Blackwell, 1953) and from partition-based bounds for stochastic programs (Song & Luedtke, 2015); Theorem R gives this monotonicity an exact form for the ceiling component of our decomposition. Second, Wasserstein distributionally robust optimization hedges against the worst-case distribution within a transport-distance ambiguity ball, typically reformulated as a convex program (Bertsimas et al., 2011; Blanchet & Murthy, 2019; Delage & Ye, 2010; Gao & Kleywegt, 2023; Mohajerin Esfahani & Kuhn, 2018); the contextual-optimization survey of Sadana et al. (2025) unifies both bridges within one literature. Theorem 1, Corollary 3, and Proposition 2 show that our minimax wave-structure selection over the deterministic elevator models coincides with this Wasserstein-DRO optimum and collapses to a closed-form rule under conditional chain dominance.

Third, robust scheduling under model uncertainty (Lu & Shen, 2021; Wiesemann et al., 2013) hedges parametric variation within a single agreed model class; our elevator-modeling uncertainty is structurally different (the throughput abstraction and true co-occupancy batching are qualitatively distinct models, not parameterizations of one family), and the Model-Dominance Hedge Rule addresses exactly this structural model-class uncertainty.

Collectively, the four problem-class threads each focus on a different facet of the wave-elevator coupling; this paper integrates the four-way interaction of wave composition, multi-story deployment, flexible AMR fleets and shared elevator capacity into a single decision problem. The first two methodological threads provide the frameworks our equivalence theorems extend, and the third positions the tools against a competing approach.

Problem Formulation

We consider a multi-story fulfillment warehouse with F floors connected by E shared freight elevators. A fleet of $|A|$ floor-bound autonomous mobile robots (AMRs) handle horizontal travel on each floor; floor-to-floor transfers go through the elevators, with each elevator carrying at most c AMRs per trip. Orders arrive over a planning horizon, each with a known source floor, destination floor, and release time. A wave is a subset of orders released into the system simultaneously at the start of a release window of length Δ . The system performance metric is the wave makespan: the time at which the last order in the wave is delivered, given a fixed operational dispatch policy.

This setting decomposes naturally into two coupled decision layers. The tactical layer chooses which orders to bundle into a wave W , namely the wave composition decision; this is the focus of our analysis. The operational layer determines, given W , which AMR serves which order, which AMRs board each elevator trip, and in what order. We delegate these decisions to a deterministic dispatch policy, defaulting to capacity-bounded FIFO boarding under the elevator model $M \in \mathcal{M}$, with full simulator details given in Section 4. We model only the wave-composition decision as a formal variable; downstream AMR-elevator execution is a simulator-realized policy. This simplification is deliberate: the wave-elevator coupling is fully captured by $\Phi(W)$ at the partition resolution on which our analytical tools operate; finer partitions reveal additional structure within Φ -cells, a property formalized as the refinement monotonicity result of Corollary 1 in Section 4. Operational lift scheduling is well studied (Chakravarty et al., 2025); the upstream wave layer that prior work treats as exogenous is formalized in this paper.

Sets and indexes

$\mathcal{O}$: set of all orders over the planning horizon, indexed by o .

$\mathcal{F} = \{1, 2, \dots, F\}$: set of floors, indexed by f .

A : AMR fleet, indexed by a .

$\mathcal{E}$: set of elevators, indexed by e .

W_ω : sequence of waves over the planning horizon, indexed by ω .

$\mathcal{M} = \{M_1, M_2, M_3\}$: elevator-model set, indexed by m .

Parameters

All parameters below are deterministic under elevator models M1 and M2. Under M3, the timing